Engineering Mechanics MEL 1010

Dynamics of particles Quiz-4

Total Time: 1 hr 45 minutes **Total Marks: 10*7=70**

Total Number of Questions: 7 [Each question carries 10 marks]

No extra time will be given. If you complete the quiz early, submit your answer sheet and leave the examination room quietly.

$a = \frac{v^2}{600} = \frac{16\times 10^{2}}{600}$

1. During a portion of a vertical loop, an airplane flies in an arc of radius $\rho = 600$ m with a constant speed $v = 400$ km/h. When the airplane is at A, the angle made by **v** with the horizontal is $\beta = 30°$, and radar tracking gives $r = 800$ m and $\theta = 30°$. Calculate v_r, v_θ, a_r, and $\ddot{\theta}$ for this instant.

$a = \frac{800}{3}$

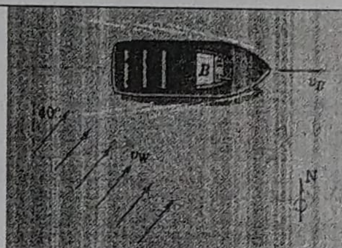
$a_\theta = r\ddot{\theta}$ $\alpha =$ $\ddot{\theta} = \frac{a_\theta}{800}$

$v_r = 200\sqrt{3}$

$v_\theta = 200$

$a_r = \frac{4\times 10^{2}}{800}$

2. A ferry is moving due east and encounters a southwest wind of speed $v_W = 10$ m/s as shown. The experienced ferry captain wishes to minimize the effects of the wind on the passengers who are on the outdoor decks. At what speed v_B should he proceed?

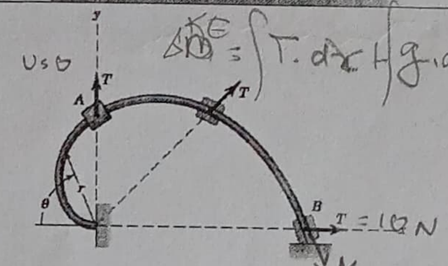


6.41

3. The 0.5-kg collar slides with negligible friction along the fixed spiral rod, which lies in the vertical plane. The rod has the shape of the spiral $r = 0.3\theta$, where r is in meters and θ is in radians. The collar is released from rest at A and slides to B under the action of a constant radial force $T = 10$ N. Calculate the velocity v of the slider as it reaches B.

$10\int_0^{\pi/2} 0.3\,d\theta + 9.8\int_0^{\pi/3} 0.3\,d\theta$

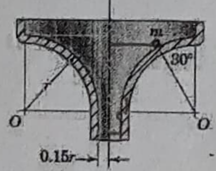
$(0.3)\left(\frac{\pi}{2}\right) = \frac{1}{2}mv^2$

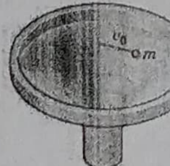


$\Delta KE = \int T\,dx + \int g\,dx$

$T = 10$ N

4. A particle is launched with a horizontal velocity $v_0 = 0.55$ m/s from the 30° position shown and then slides without friction along the funnel-like surface. Determine the angle θ which its velocity vector makes with the horizontal as the particle passes level O-O. The value of r is 0.9 m.



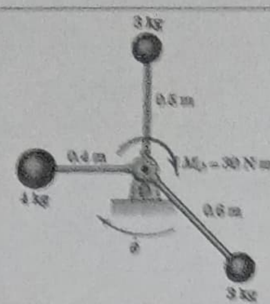
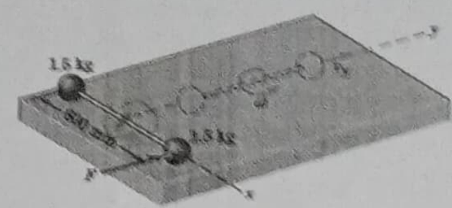
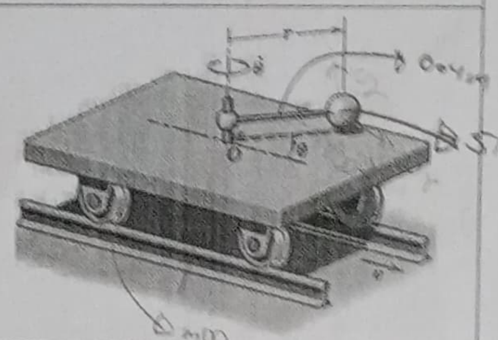


$\dot{\theta} r e_\theta + \dot{\theta} r(-\theta e_r) + \dot{\theta}\dot{r} e_\theta$

$\dot{\theta} r = 55.56$ $+ \; \dot{r}(e_\theta)(\dot{\theta}) + \ddot{r} e_r$

$(e_r)(\ddot{r} - \dot{\theta}^2 r)$

$a =$ $+ (e_\theta)(\ddot{\theta} r + 2\dot{\theta}\dot{r})$

5. The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20$ rad/s. If a couple $M_o = 30$ N.m is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$.	
6. The two spheres are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y-direction and imparts an impulse of 10 Ns. during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.	
7. The small car, which has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\dot{\theta} = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0$, calculate v when $\theta = 60°$. Neglect the mass of wheels and any friction.	

$\frac{1}{2}\times 25\times(0.6)^2 + \frac{1}{2}(5\times 0.4^2)\,16 =$

CG

$a = \omega^2 R$
$= 16\times 0.4$
$= 6.4$ m/s²

$a\cos\theta$

$a = -6.4\cos\theta$

$\frac{da}{dt} = +6.4\sin\theta \times 4$

$\frac{da}{dt} = 25.6\sin\theta$

$\left(\frac{a}{a}\ \frac{b}{b}\ \frac{c}{2c}\right)$

$\left(1\ \ 1\ \ \frac{1}{2}\right)$

$(1\ \ 1\ \ 2)$

y, b, a, c/2, z

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Indian Institute of Technology Jodhpur
NH 65, Surpura Bypass Rd, Karwar, Rajasthan 342037

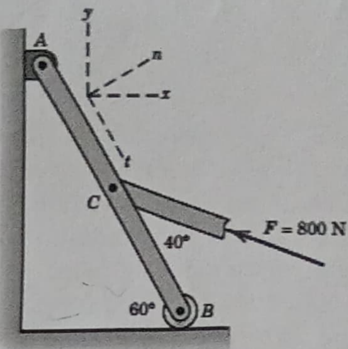
Minor Exam
Date: 22/02/2025

MEL 1010: Engineering Mechanics

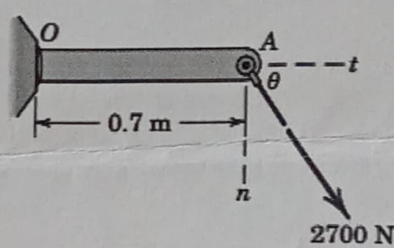
Maximum marks: 40

Time: 120 mins

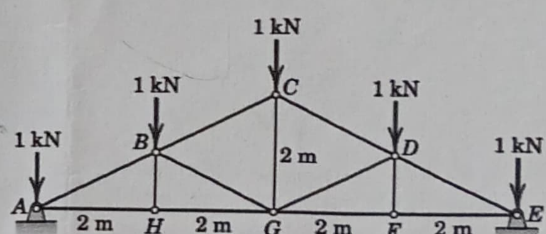
Q1: A force **F** of magnitude 800 N is applied to point C of the bar AB as shown. Determine both the x-y and the n-t components of **F**. **[4 marks]**



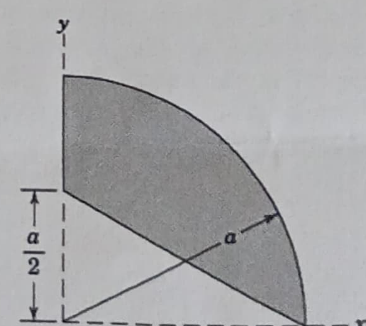
Q2: The weld at O can support a maximum of 2500 N of force along each of the n- and t-directions and a maximum of 1400 N-m of moment. Determine the allowable range for the direction θ of the 2700-N force applied at A. The angle θ is restricted to $0 \leq \theta \leq 90°$. **[3 marks]**



Q3: Neglect any horizontal reactions at the supports and solve for the forces in all members. **[6 marks]**



Q4: Determine the x- and y-coordinates of the shaded area. Use the equation of a circle for the curve with "a" as radius. **[7 marks]**



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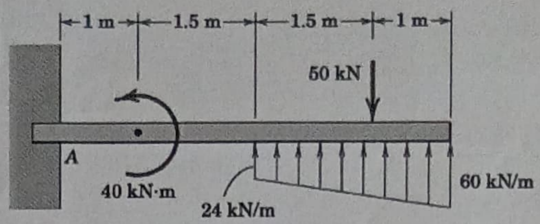
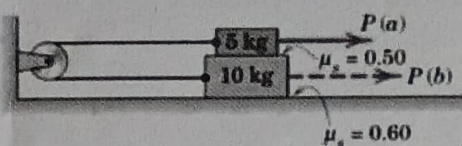
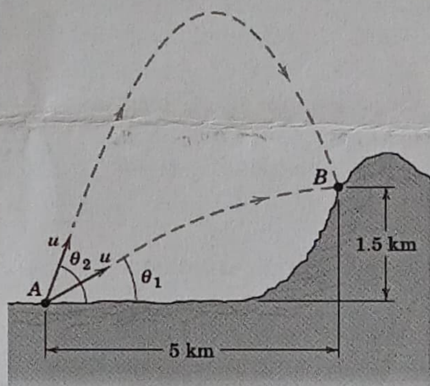
Q5: Determine the force and moment reactions at A for the beam which is subjected to the load combination shown. **[5 marks]**



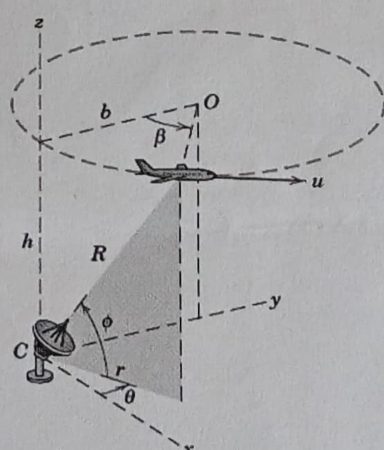
Q6: The system of two blocks, cable, and fixed pulley is initially at rest. Determine the horizontal force P necessary to cause motion when (a) P is applied to the 5-kg block and (b) P is applied to the 10-kg block. Determine the corresponding tension T in the cable for each case. **[5 marks]**



Q7: The muzzle velocity of a long-range rifle with u=400 m/s at A is. Determine the two angles of elevation which will permit the projectile to hit the mountain target B. **[4 marks]**



Q8: An aircraft is flying in a horizontal circle of radius b with a constant speed u at an altitude h. A radar tracking unit is located at C. Write expressions for the components of the velocity of the aircraft in the spherical coordinates of the radar station for a given position.
[6 marks]



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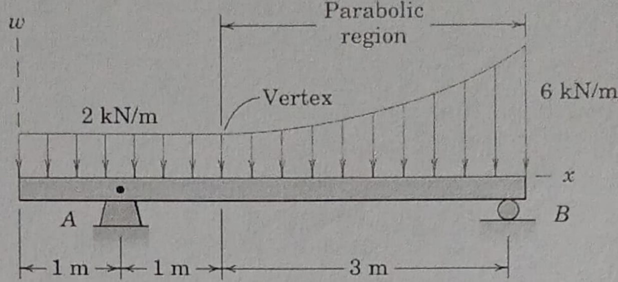
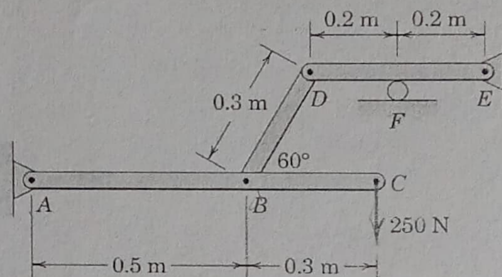
Course: Engineering Mechanics (Major Exam)
Code: MEL1010
Date: 27/04/2025 **Duration : 180 mins** **Max marks: 80**

Question 1: Determine the reactions at the supports of the beam which is acted on by the combination of uniform and parabolic loading distributions **[10 marks]**

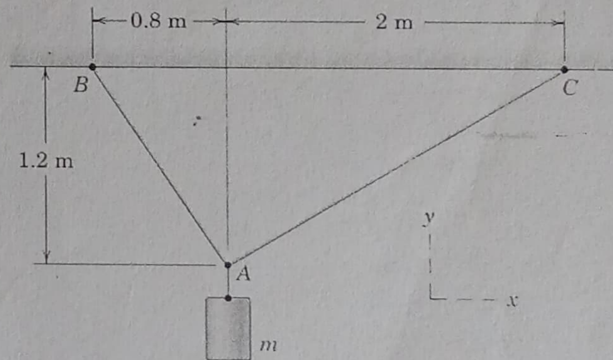


Question 2: Determine the reaction at the roller F for the frame loaded as shown.

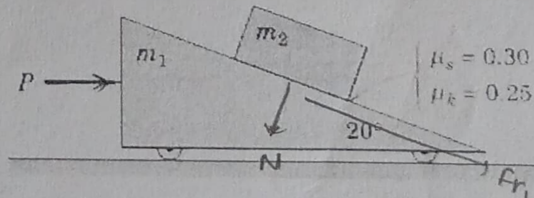
[5 marks]



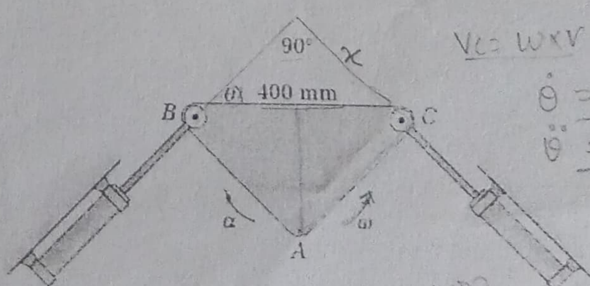
Question 3: Write each tension force acting on point A as a vector if the mass m is 60 kg. **[5 marks]**



Question 4: Determine the applied force P for which the mass m_2 doesn't slip on the wedge shape mass m_1. Find the range of P in terms of m_1 and m_2. **[15 marks]**



Question 5: At the instant represented $\theta = 45°$ and the triangular plate ABC has a counter clockwise angular velocity of 20 rad/s and a clockwise angular acceleration of 100 rad/s^2. Determine the



$V_C = \omega \times r$

$\dot{\theta} = \omega$

$\ddot{\theta} = \alpha$

$\frac{x}{\sin\theta} = \frac{400}{\sin 90}$

$x = 400 \sin\theta$

$\dot{x} = 400 \cos\theta \, \dot{\theta}$

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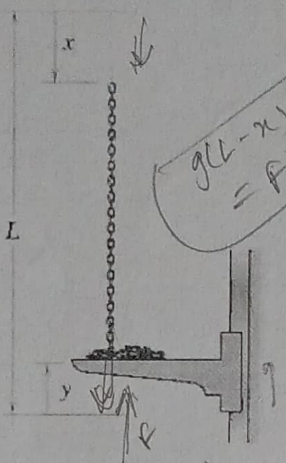
magnitudes of the corresponding velocity **v** and acceleration **a** of the piston rod of the hydraulic cylinder attached to C. **[10 marks]**

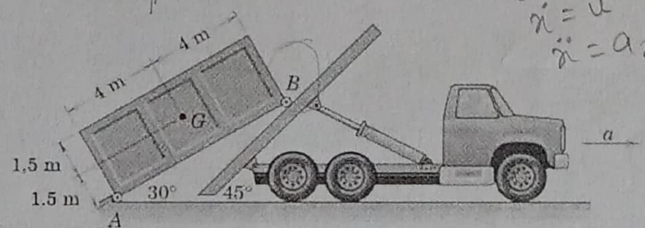
Question 6: The open-link chain of total length L and of mass ρ per unit length is released from rest at x = 0 at the same instant that the platform starts from rest at y=0 and moves vertically up with a constant acceleration a. Determine the expression for the total force exerted on the platform by the chain t seconds after the motion starts. **[15 marks]**



$g(L-x)\rho + \rho x(a+g) = F$

$\ddot{y} = a$ $\rho x(\dot{x} + a)$
$\dot{x} = u$
$\ddot{x} = a_x$

Question 7: The figure shows a roll-off truck ramp for discharging loaded containers. The loaded 120000 kg container may be treated as a homogeneous solid rectangular block with mass center at G. If the supporting wheel A is restrained from movement, calculate the force F_B exerted by the ramp on the supporting wheel B when the truck starts frnm rest with a forward acceleration of 3 m/s^2. Neglect friction at B. **[10 marks]**



$\rho(L-x)\frac{v\,du}{dx} + x\rho(g+a)$

Question 8: The four-bladed fan rotates about the fixed axis OB with a constant angular speed N = 1200 rev/min. Write the vector expressions for the velocity v and ac- celeration a of the tip A of the fan blade for the in- st ant when its x-y-z coordinates are 0.260, 0.240, and 0.473 m, respectively. **[10 marks]**